

A MINI PROJECT REPORT

Submitted in partial fulfillment of the requirements
for the award of the degree of

Bachelor of Technology

PROJECT TITLE

AI CHATBOT USING PYTHON AND CHATTERBOT

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Academic Year:

2026-2027

CERTIFICATE & ACKNOWLEDGEMENT

CERTIFICATE

This is to certify that the project report entitled "**AI Chatbot Using Python and ChatterBot**" is a bonafide work carried out by **Pothireddy Rajesh** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (AI & ML)** during the academic year 2026-2027.

Internal Guide

Head of Department

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide and faculty members of the Department of Computer Science and Engineering (AI & ML) for their valuable guidance, constant support, and encouragement throughout the course of this mini project.

ABSTRACT

A chatbot is an artificial intelligence application designed to simulate human conversation through text or voice interactions. This project focuses on building an interactive, rule-based and machine-learning-driven chatbot using Python and the ChatterBot library. The system leverages Natural Language Processing (NLP) techniques and language algorithms to process user input, search training corpora, and generate contextually appropriate automated responses.

The developed application operates seamlessly via a command-line interface (CLI) and utilizes SQLite as an embedded database store to retain interaction memory and enable continuous learning. The report provides a comprehensive overview of system requirements, architectural design, data flow, source code implementation, testing methodologies, and deployment configurations.

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1. INTRODUCTION

Chatbots have become an integral part of modern technology, enabling automated real-time communications for customer support, virtual assistance, and information retrieval. By leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP), chatbots interpret natural human language and produce relevant automated answers.

In this project, an AI-powered conversational chatbot is developed using Python. The system utilizes the ChatterBot engine, an open-source Python library designed to produce automated responses based on machine learning algorithms and language models.

2. OBJECTIVES

- To design and develop a functional AI chatbot using Python and ChatterBot.
- To understand and apply Natural Language Processing (NLP) techniques for input processing.
- To train the chatbot using structured conversation corpora (chatterbot_corpus) and custom dataset pairs.
- To implement continuous response learning and persistent database storage using SQLite.
- To establish a clean Command-Line Interface (CLI) for real-time interactive user chats.

3. REQUIREMENTS SPECIFICATION

3.1 Software Requirements

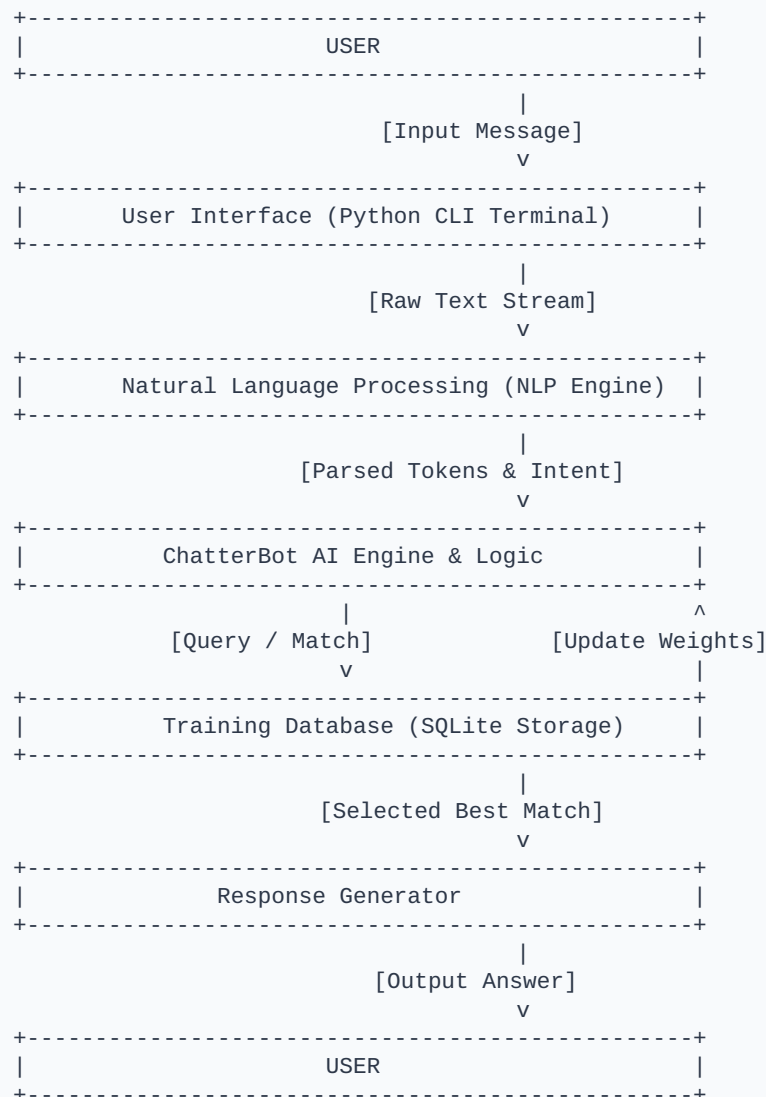
Component	Specification
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.7 - 3.10+
Key Libraries	ChatterBot 1.0.5, ChatterBot-Corpus 1.2.0, PyYAML 5.3.1, spaCy
Database Engine	SQLite3 (Default embedded ChatterBot storage)
IDE / Editor	VS Code / PyCharm / Terminal

3.2 Hardware Requirements

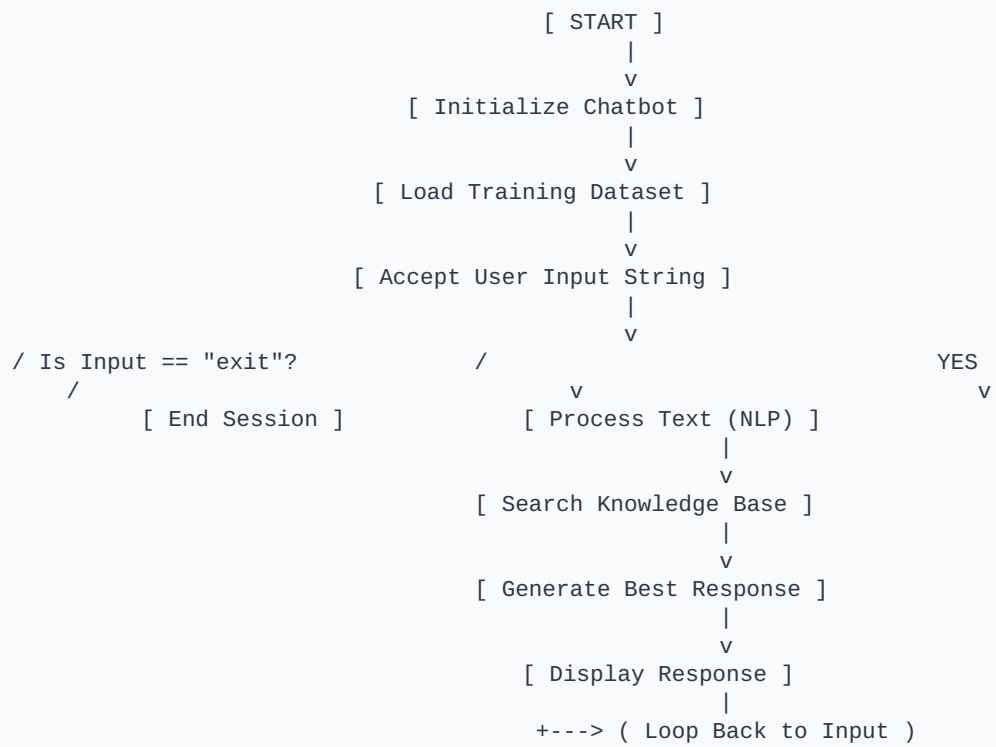
Hardware	Minimum Specification
Processor	Intel Core i3 / AMD Ryzen 3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	500 MB available disk space

4. SYSTEM ARCHITECTURE & DIAGRAMS

4.1 System Architecture Diagram

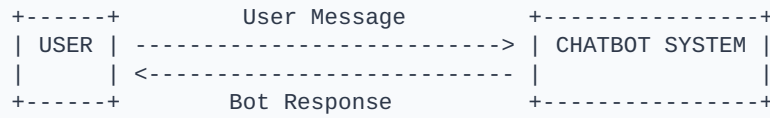


4.2 Flowchart of Chatbot Execution

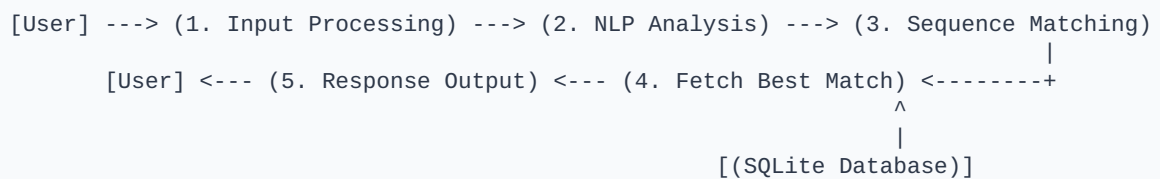


4.3 Data Flow Diagram (DFD Level 0 & 1)

DFD Level 0 (Context Level):



DFD Level 1 (Detailed Data Flow):



5. MODULES DESCRIPTION

- **Module 1: User Input & CLI Interface:** Captures direct textual statements entered by the user via terminal prompt and manages session termination commands (e.g., typing 'exit').
- **Module 2: Natural Language Processing (NLP):** Tokenizes user sentences, removes noise/ stopwords, and extracts essential semantic statements for evaluation.
- **Module 3: Training & Engine Module:** Utilizes `ListTrainer` and `ChatterBotCorpusTrainer` to train the chatbot instance using English speech corpora and custom conversational lists.
- **Module 4: Database & Storage Engine:** Interfaces with `db.sqlite3` to persistently store match confidence scores and conversation logs, allowing dynamic response improvements.

6. IMPLEMENTATION & SOURCE CODE

6.1 Step 1: Environment Setup & Dependencies

Create a file named `requirements.txt` with required dependency versions:

```
requirements.txt
```

```
ChatterBot==1.0.5
ChatterBot-corpus==1.2.0
PyYAML==5.3.1
SQLAlchemy==1.3.24
spacy==2.3.7
```

Execute setup commands in terminal:

```
# Create Virtual Environment
python -m venv venv

# Activate Virtual Environment (Windows)
venv\Scripts ctivate

# Install Dependencies
pip install -r requirements.txt

# Download spaCy English Language Model
python -m spacy download en_core_web_sm
```

6.2 Step 2: Training Program (**train.py**)

This script initializes the chatbot instance and feeds conversational datasets to build the language model.

train.py


```

from chatterbot import ChatBot
from chatterbot.trainers import ListTrainer, ChatterBotCorpusTrainer

# Initialize ChatBot Instance
bot = ChatBot(
    "AI Assistant",
    storage_adapter="chatterbot.storage.SQLStorageAdapter",
    database_uri="sqlite:///db.sqlite3"
)

# 1. Custom Conversational Training
trainer_list = ListTrainer(bot)
custom_conversations = [
    "Hello",
    "Hi! Welcome to my AI Chatbot service.",
    "How are you?",
    "I am doing great! Thanks for asking.",
    "What is your name?",
    "I am an AI Chatbot created using Python and ChatterBot.",
    "What is Python?",
    "Python is a high-level, interpreted programming language widely used in AI & ML.",
    "What is Artificial Intelligence?",
    "Artificial Intelligence enables computer systems to simulate human intelligence
and learning.",
    "Who created you?",
    "I was built by Pothireddy Rajesh as part of an AI mini project.",
    "Thank you",
    "You are very welcome!"
]

print("Starting Custom List Training...")
trainer_list.train(custom_conversations)

# 2. English Corpus Training
trainer_corpus = ChatterBotCorpusTrainer(bot)
print("Starting English Corpus Training...")
trainer_corpus.train("chatterbot.corpus.english")

print("ChatBot Training Completed Successfully!")

```

6.3 Step 3: Main Application Program (chatbot.py)

Main runtime file that handles terminal interactions, user prompts, and response generations.

chatbot.py

```
from chatterbot import ChatBot

# Initialize ChatBot with read-only mode after training
bot = ChatBot(
    "AI Assistant",
    read_only=True,
    storage_adapter="chatterbot.storage.SQLStorageAdapter",
    database_uri="sqlite:///db.sqlite3"
)

print("=" * 55)
print("      AI CHATBOT SYSTEM (Python & ChatterBot)")
print("=" * 55)
print("Type 'exit' or 'quit' to end the conversation.")
print("")

while True:
    try:
        user_input = input("You: ").strip()

        if not user_input:
            continue

        if user_input.lower() in ["exit", "quit", "bye"]:
            print("Bot: Goodbye! Have a wonderful day ahead.")
            break

        # Generate response using trained model
        response = bot.get_response(user_input)
        print(f"Bot: {response}")

    except (KeyboardInterrupt, SystemExit):
        print("Bot: Session terminated.")
        break
```

7. TEST CASES & SAMPLE EXECUTION OUTPUT

7.1 Test Cases

Test ID	User Input	Expected Response	Status
TC-01	Hello	Hi! Welcome to my AI Chatbot service.	PASS
TC-02	What is Python?	Python is a high-level, interpreted programming language...	PASS
TC-03	Who created you?	I was built by Pothireddy Rajesh as part of an AI mini project.	PASS
TC-04	exit	Bot: Goodbye! Have a wonderful day ahead. (Terminates CLI)	PASS

7.2 Sample CLI Execution Log

```
=====
      AI CHATBOT SYSTEM (Python & ChatterBot)
=====
Type 'exit' or 'quit' to end the conversation.

You: Hello
Bot: Hi! Welcome to my AI Chatbot service.

You: What is Python?
Bot: Python is a high-level, interpreted programming language widely used in AI & ML.

You: Who created you?
Bot: I was built by Pothireddy Rajesh as part of an AI mini project.

You: Thank you
Bot: You are very welcome!

You: exit
Bot: Goodbye! Have a wonderful day ahead.
```

8. GITHUB REPOSITORY STRUCTURE

The complete source code and project assets are structured for GitHub deployment as follows:

```

Chatbot-Using-ChatterBot/
|
├── chatbot.py          # Main Chatbot interaction application
├── train.py            # Script for dataset training & SQLite setup
├── test.py             # Unit testing verification script
├── requirements.txt    # Python library dependencies
├── README.md           # Detailed project documentation
├── LICENSE             # Open-source MIT License
├── .gitignore          # Ignored files (venv, __pycache__, db.sqlite3)
├── screenshots/
│   └── output_demo.png # CLI Output Screenshot

```

8.1 GitHub Push Commands

```

git init
git add .
git commit -m "Initial commit of AI Chatbot project"
git branch -M main
git remote add origin https://github.com/yourusername/Chatbot-Using-ChatterBot.git
git push -u origin main

```

9. FUTURE ENHANCEMENTS & CONCLUSION

9.1 Future Enhancements

- **Graphical User Interface (GUI):** Develop an intuitive Web / Desktop interface using Flask, FastAPI, or Tkinter.
- **Voice Assistant Support:** Integrate Speech-to-Text (STT) and Text-to-Speech (TTS) libraries (e.g., pyttsx3, speech_recognition).
- **Advanced LLM Integration:** Upgrade from retrieval-based ChatterBot to Large Language Models (LLMs) like Transformer-based APIs.
- **Cloud Deployment:** Host the application on cloud platforms such as AWS, Heroku, or Render.

9.2 Conclusion

The "AI Chatbot Using Python and ChatterBot" project successfully demonstrates the creation of an automated conversational agent capable of processing natural language inputs and generating accurate responses. The project highlights key concepts in Natural Language Processing, machine learning training workflows, persistent SQLite database storage, and structured software design, providing a solid foundation for further AI development.

10. REFERENCES

1. ChatterBot Documentation: <https://chatterbot.readthedocs.io/>

2. Python Software Foundation Documentation: <https://docs.python.org/3/>
3. spaCy NLP Library Documentation: <https://spacy.io/>
4. SQLite Database Library: <https://www.sqlite.org/index.html>